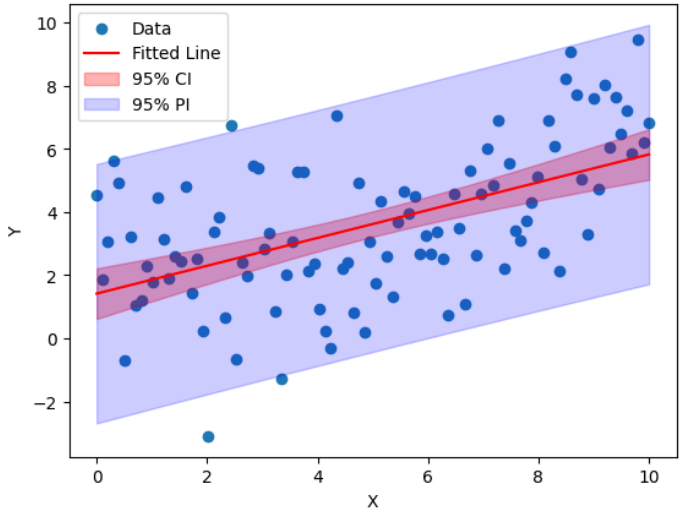
Lecture 24

Uncertainty Quantification II



Announcements

- Course project feedback out tonight
  - Directed questions out later this week

- Office hours every Tuesday (noon to 4pm) and Friday (11am to 1pm)
  - Link to book appointments is on the course website

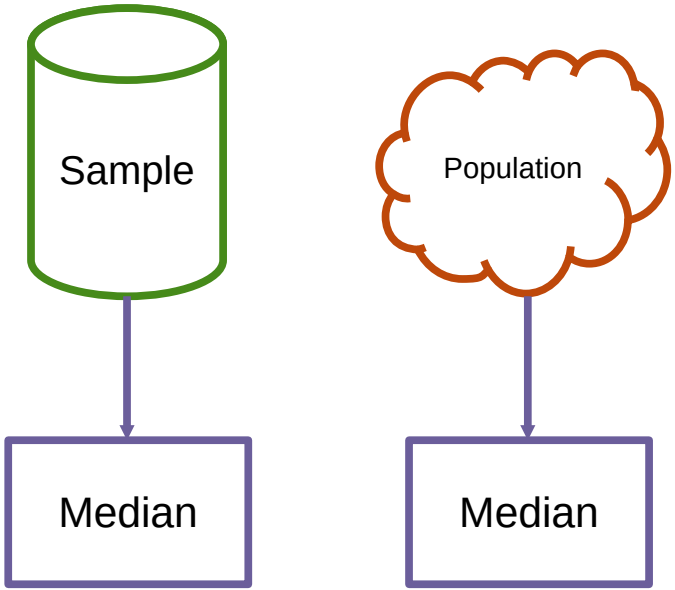
Today

Confidence Intervals

- Idea
- Bootstrapping

Statistics

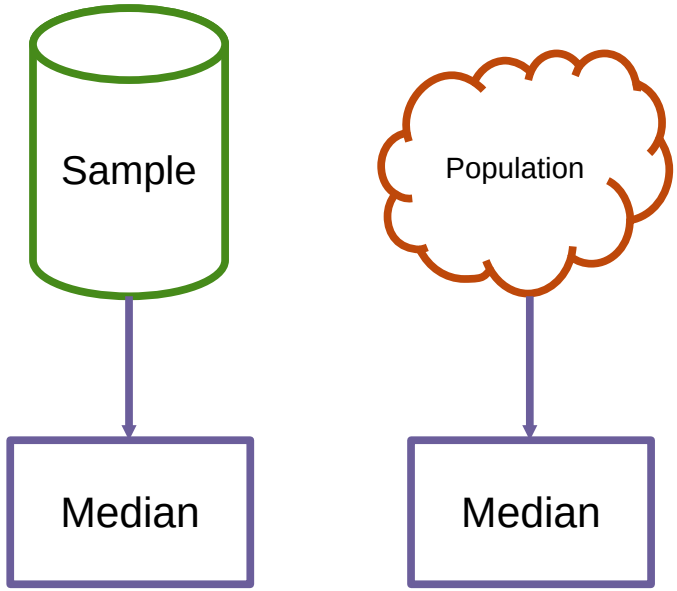
With inference, we try to learn about population statistics from sample statistics



Statistics

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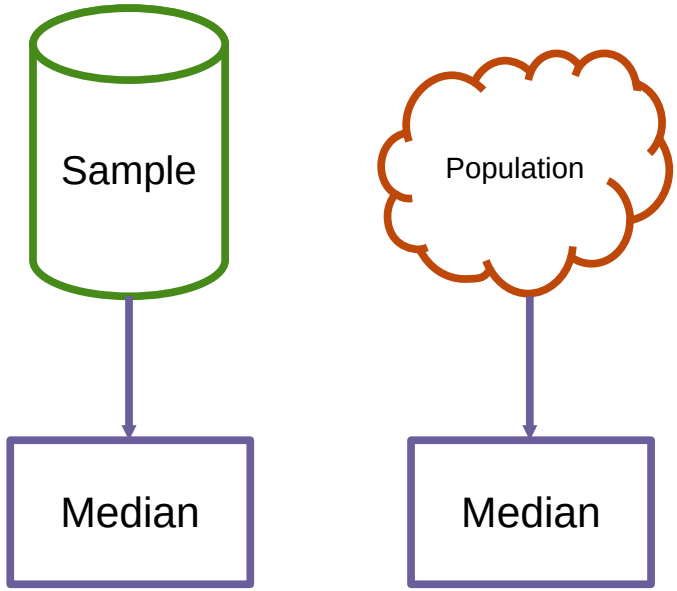
- Due to sampling variance, the median of the sample may be different from the population
- Even if the sampling is unbiased (sample statistic true on average)



Statistics

With inference, we try to learn about population statistics from sample statistics

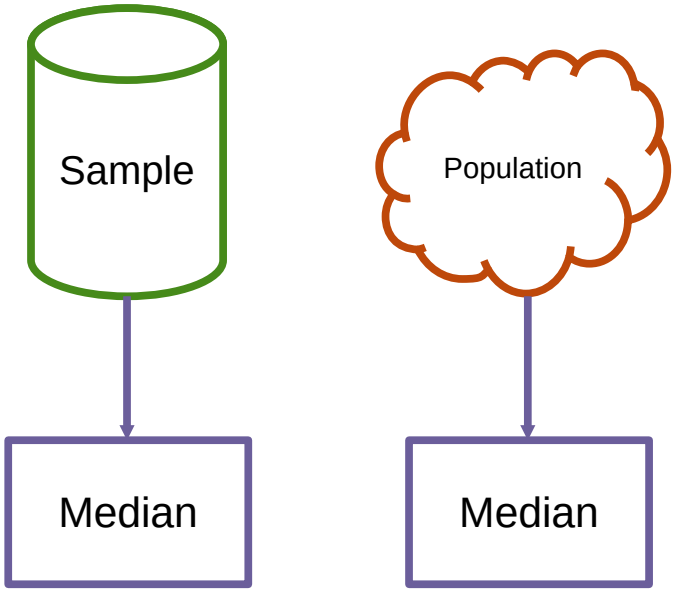
- Due to sampling variance, the median of the sample may be different from the population
- Even if the sampling is unbiased (sample statistic true on average)
- From the sample, let's infer a range of values that we think the population median may take (**confidence interval**)



How much do students walk each week?

Suppose we want to know how many miles Allegheny students walk each week

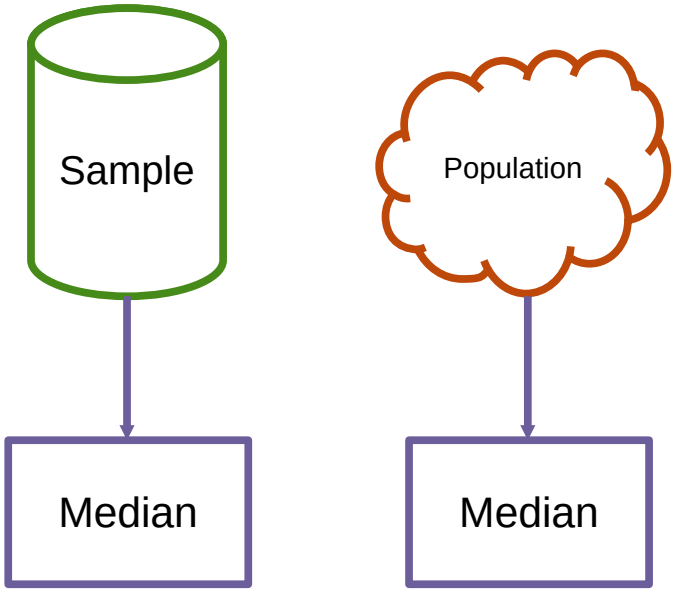
- There is a population median
- Collect data from 50 random students



How much do students walk each week?

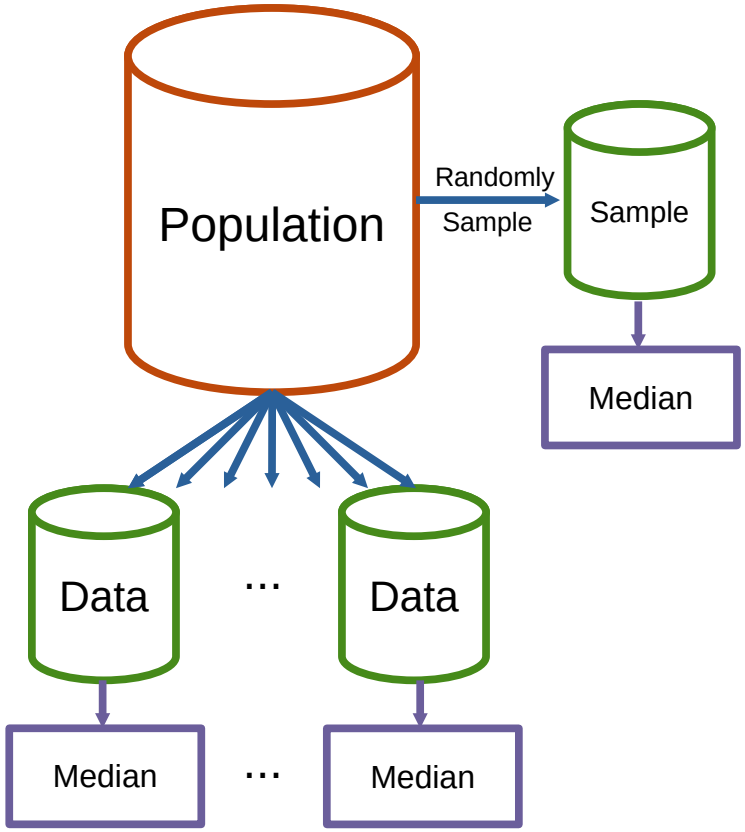
Suppose we want to know how many miles Allegheny students walk each week

- There is a population median
- Collect data from 50 random students
- We can use this sample to build a confidence interval for the population median



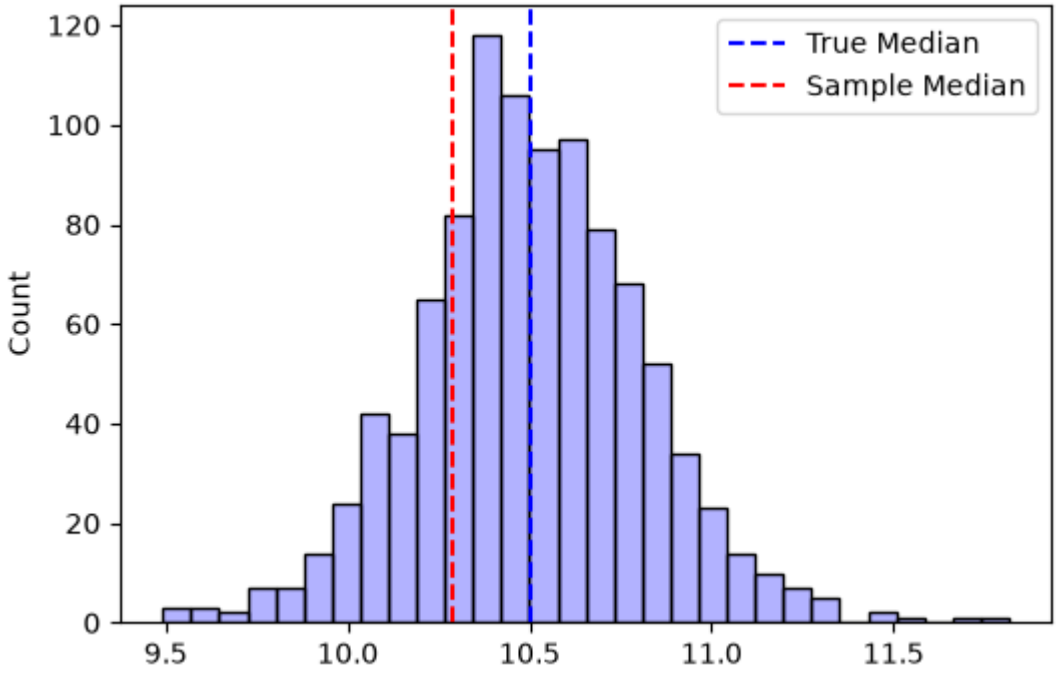
Sampling Variance

Recall that if we had access to the population, we could measure sampling variance



Sampling Variance

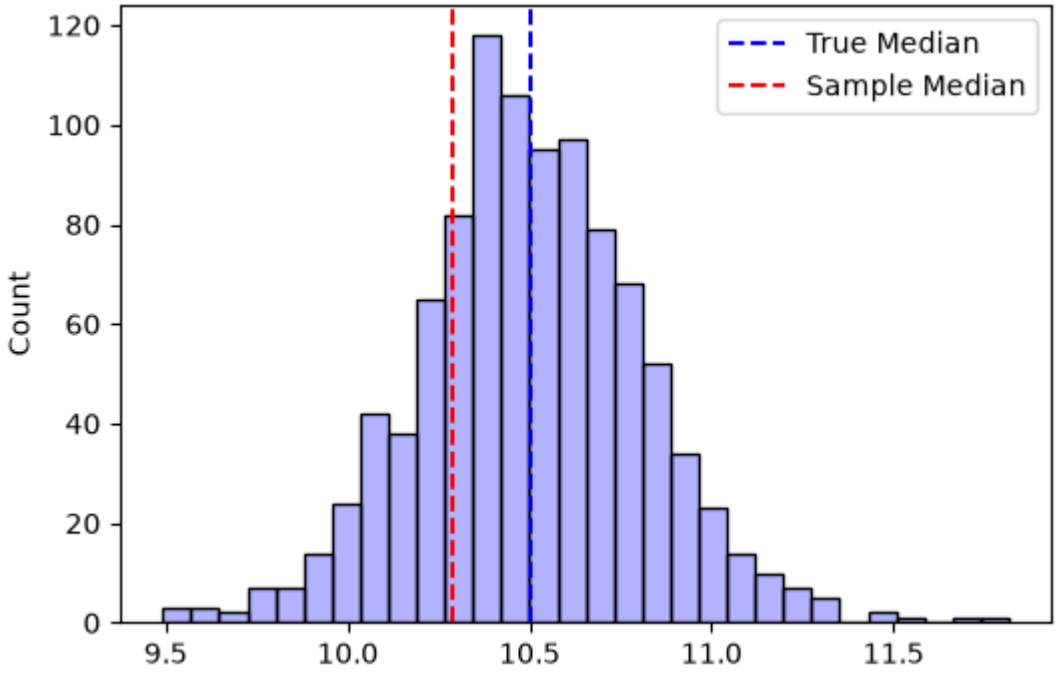
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Sampling Variance

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- Unrealistic assumption



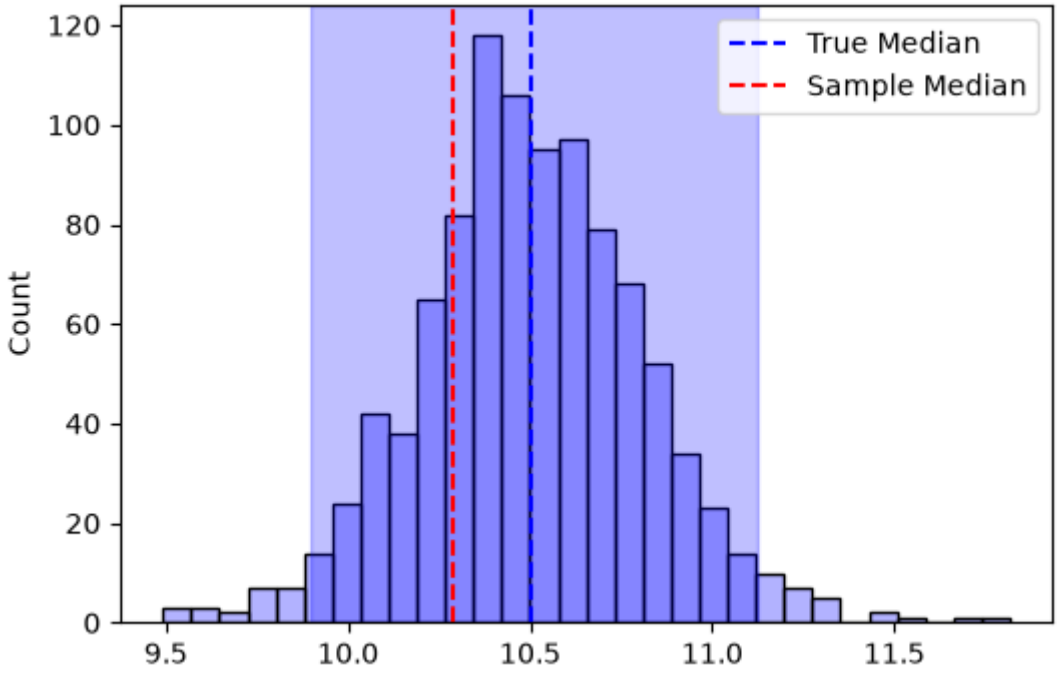
Sampling Variance

Recall that if we had access to the population, we could measure sampling variance

- Unrealistic assumption

Obtain a range for the sample median by taking the middle 95%

- 95% of the time, the sample falls between 9.9 and 11.1

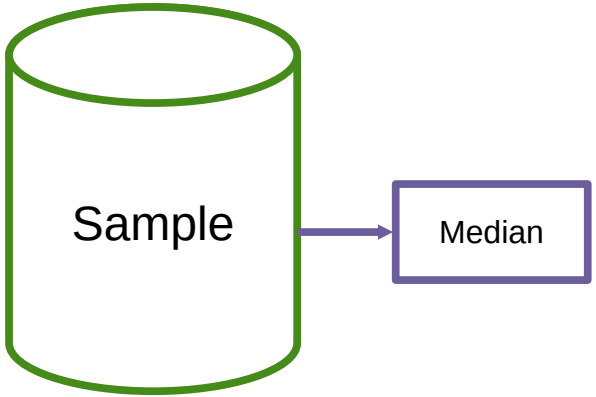


Bootstrapping

We don't have access to the population!

Let's treat our sample as "representative" of our population and resample from it

- Bootstrapping

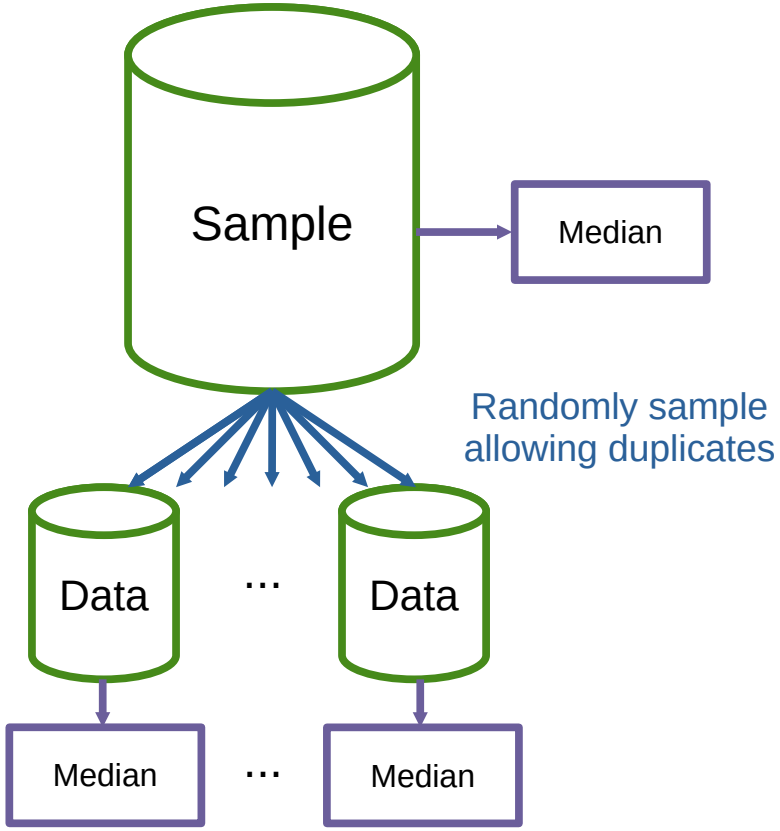


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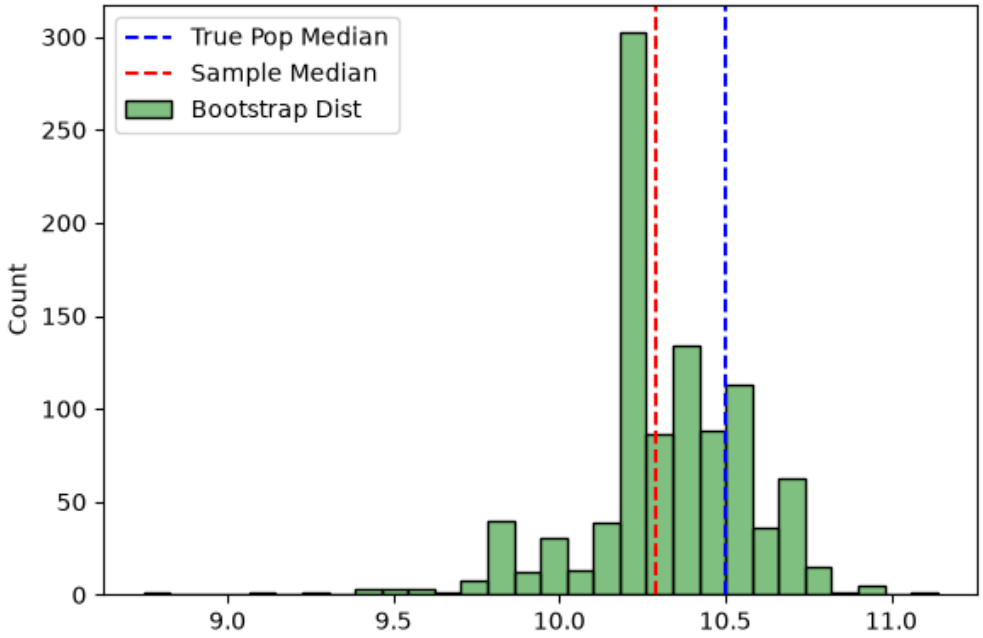


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- Gives us a distribution of medians

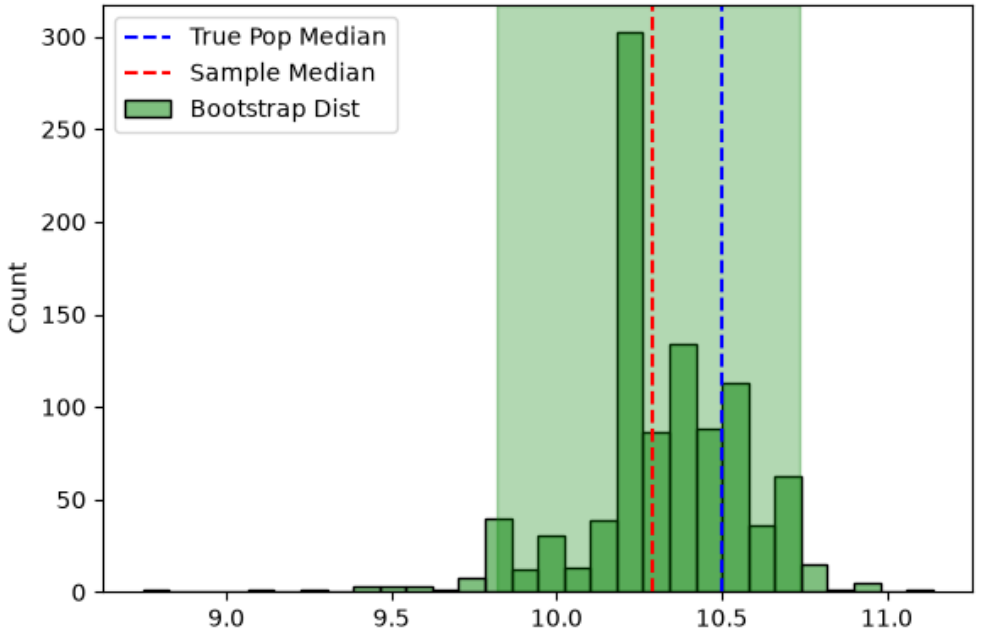


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- Bootstrapping
- Gives us a distribution of medians
- Taking the middle 95% gives us a range of values for the unknown population median

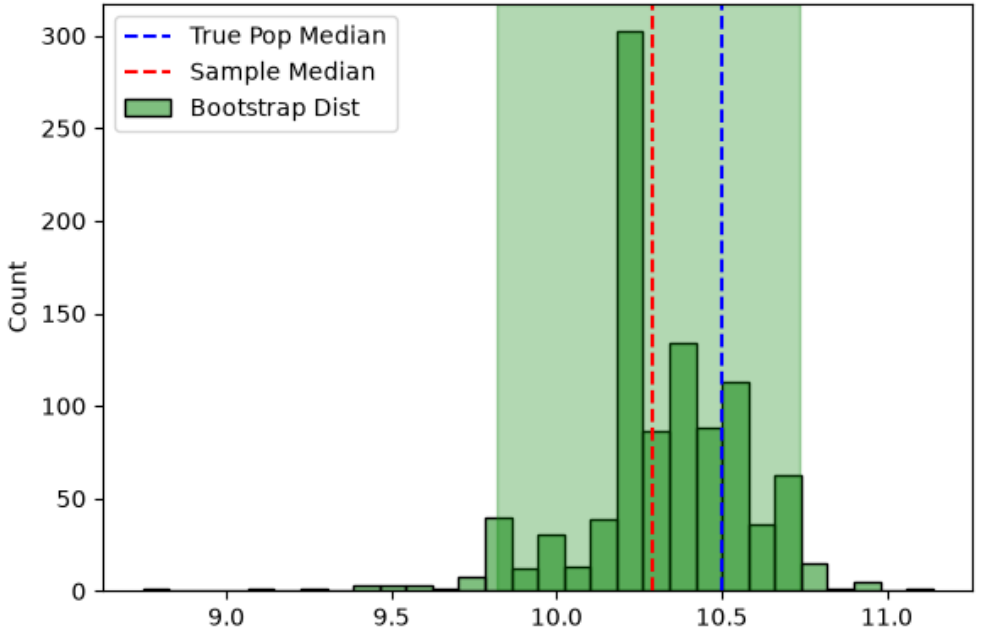


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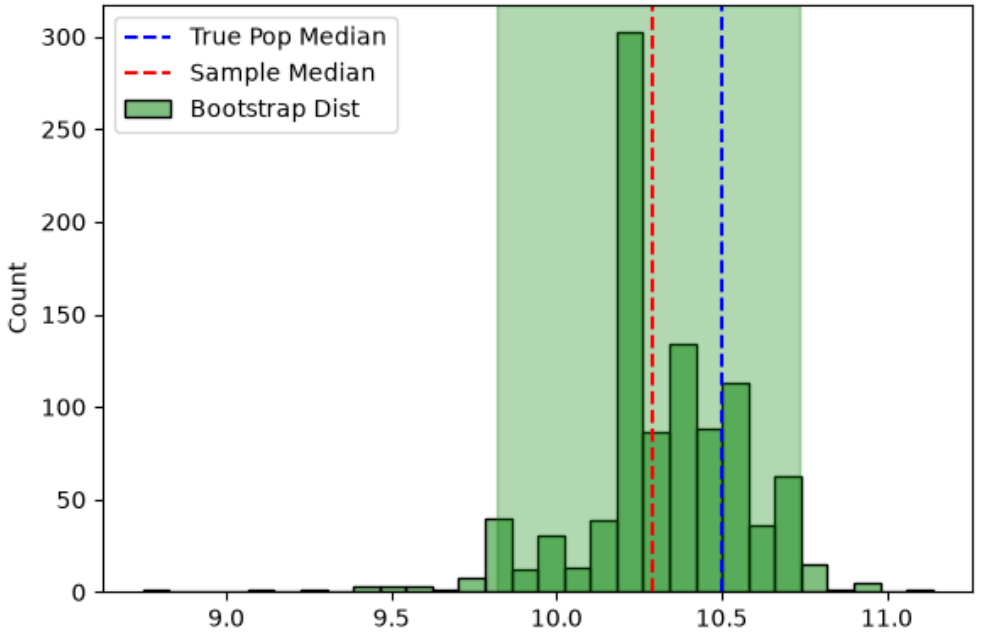
Let's treat our sample as "representative" of our population and resample from it

- Bootstrapping
- Gives us a distribution of medians
- Taking the middle 95% gives us a range of values for the unknown population median
- "We think the population median is between 9.8 and 10.7"



Interpretation

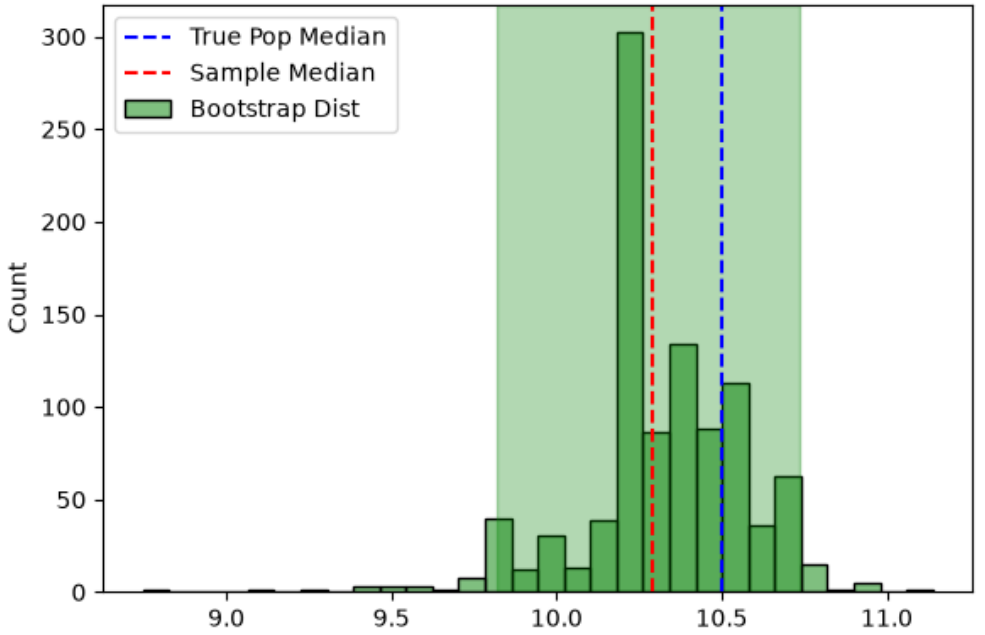
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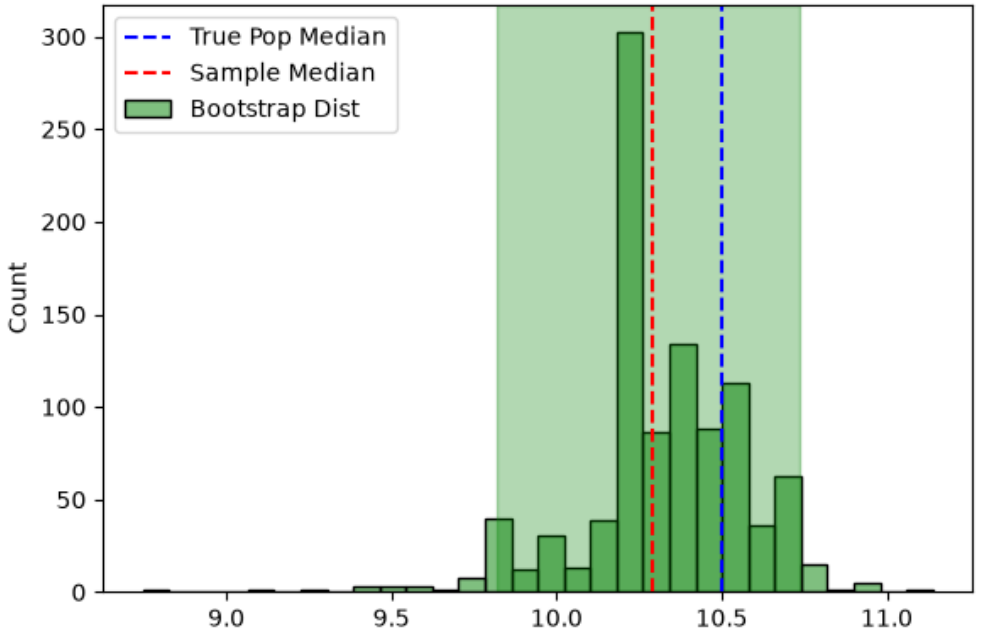


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If we obtained new samples and computed bootstrapped confidence intervals, then the intervals would contain the population median 95% of the time



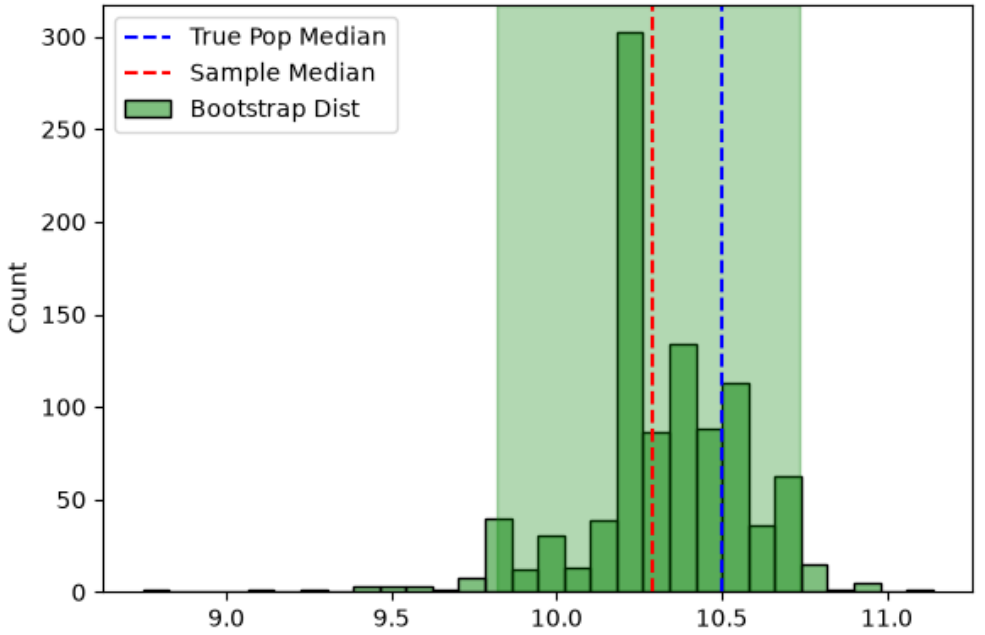
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If we obtained new samples and computed bootstrapped confidence intervals, then the intervals would contain the population median 95% of the time

- Nevertheless, we call the shaded green region a 95% confidence interval for the population median



Next Week

- Linear relationships